September 14

random events: outcome cannot be predicted with certainty

- we DO know the possible outcomes

each outcome has a probability, which quantifies the relative likelihood of that outcome occurring

$\mathbb{P}$(outcome A) = relative likelihood of A

it's typical to think of this as the long-run frequency of that outcome, if the random event were repeated many times.

eg flipping a coin - $\mathbb{P}(\text{heads}) = 0.5$ says if we flipped the coin many, many times, it would come up heads ½ of the time.

convention: probabilities are between 0 and 1 (can be expressed as a decimal, fraction, or percentage)

$0 \leq \mathbb{P}(\text{outcome A}) \leq 1$

probability of 0 means that outcome will NOT occur

$\mathbb{P}(\text{rain tomorrow}) = 0$ → definitely not rain

probability of 1 means that outcome definitely will occur

$\mathbb{P}(\text{rain tomorrow}) = 1$ → definitely rain

another convention: the probabilities of all the different outcomes add up to 1

$\mathbb{P}(\text{rain}) + \mathbb{P}(\text{no rain}) = 1$

$\mathbb{P}(\text{rain}) = 70\% \Rightarrow \mathbb{P}(\text{no rain}) = 30\%$

The result of the random event is a "random variable"

random variable	possible values
weather tomorrow	rain, no rain
which side of flipped coin lands facing up	heads, tails

Those are random variables with 2 possible outcomes.

We can also have RVs that are categorical...

	RV	values
categorical	color of M&M drawn from bag	blue, green, red

... and RVs that take numerical values

	RV	values
discrete	# of red M+Ms when you draw 3 from bag	0, 1, 2, 3

↑

This is an example of a "discrete" random variable - the outcomes are integers that we can list out.

There are also "continuous" RVs that can take any value in some interval. For instance, a patient's blood pressure (systolic)

	RV	values
continuous	blood pressure	(70, 200)

The distribution of a random variable is the set of all its possible values, and their corresponding probabilities.

If X = result of coin flip, the distribution of X is:

x	heads	tails
P(X)	0.5	0.5

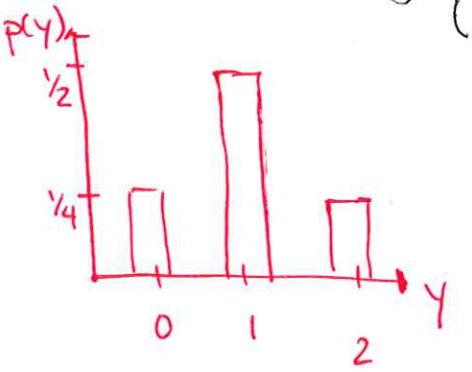
The # of tails in 2 coin flips is also a random variable

Y = # tails

y	0	1	2
$p(y)$	1/4	1/2	1/4

"probability mass function" (pmf)

The possible values are 0, 1, 2, and the probabilities of each are 1/4, 1/2, and 1/4 respectively. (for a fair coin)



We could also represent that information visually.

For something continuous, eg blood pressure, we don't have a probability of observing an *exact value*, like 110, because the variable can take any value in an interval, and all the probabilities have to sum to 1.

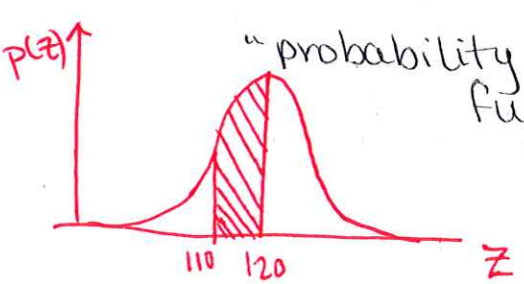
Z continuous

$\mathbb{P}(Z = a) = 0$

$\mathbb{P}(a < Z < b) > 0 \quad (a \neq b)$

Instead, the distribution of a continuous RV assigns probability of the RV falling between 2 different values.

Instead of the histogram/bargraph for the coin flip example, the continuous distribution is visually represented by a curve, and the probability of Z falling between 2 values is given by the corresponding area under the curve.



area = $\mathbb{P}(110 < Z < 120)$

There are several important quantities we use to summarize the distrib. of a RV. One of those is the expected value.

expected value = weighted avg of outcomes

↳ it tells us what a "typical" value is for this variable

↳ and is defined as a weighted avg of the possible outcomes, weighted by how likely that outcome is to occur.

So if our RV is Y = # tails in 2 coin flips, Y = 0, 1, 2

then $E(Y) = 0 \cdot P(Y=0) + 1 \cdot P(Y=1) + 2 \cdot P(Y=2)$

$= 0 \cdot 1/4 + 1 \cdot 1/2 + 2 \cdot 1/4$

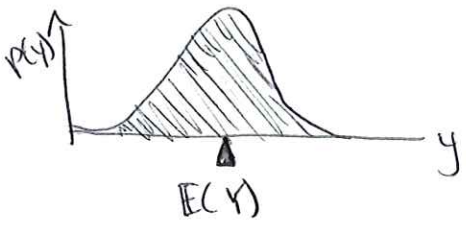
$= 1$

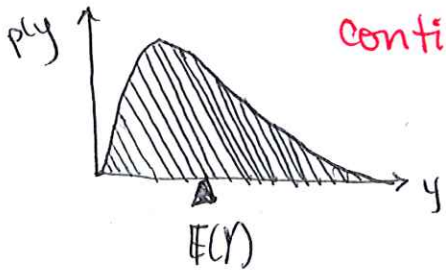
(aside: This is for RVs that take numerical values! Does not make sense for categorical variables.)

more generally: $E(Y) = \sum_{\text{all values of } y} y \cdot P(Y=y)$

For a continuous RV, it's the same idea: still a probability-weighted avg of the possible outcome values, but you can't add up that sum directly anymore.

you can think of expected value as the "balancing point" of the distribution: imagine we literally cut out a piece of wood in the shape of the prob. density function, and put a fulcrum under it.

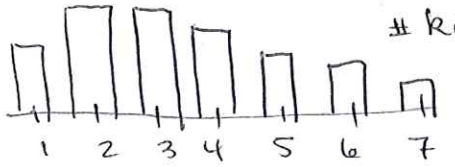


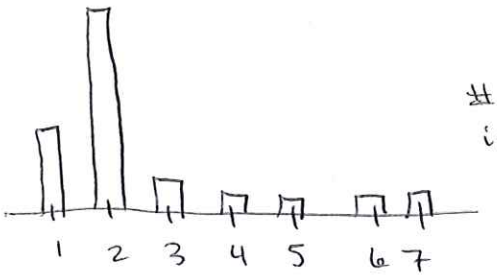


continuous Y → E(Y) is "balancing point" of pdf

Another important quantity we use to describe the probability distribution of a RV is its variance

variance - how spread out the values of this RV tend to be





For the same range of possible outcome values, a RV can be more spread out, or more concentrated.

How do we quantify this?

$$\text{var}(X) = \mathbb{E}\left(\{X - \mathbb{E}(X)\}^2\right)$$

$$= \sum_x (x - \mathbb{E}(X))^2 \mathbb{P}(X = x)$$

(aka σ_x^2)

it's the expected value of the squared distance from the distribution's expected value

Why does this tell us how spread out it is?

If the variable takes values that are all close together, then $x - \mathbb{E}(X)$ is small, and $(x - \mathbb{E}(X))^2$ is small. If the most likely values of X are all close together, then $\mathbb{P}(X = x)$ is big for values of $(x - \mathbb{E}(X))^2$ that are small.

aside:

Why $(x - E(X))^2$? Why do we square the distances?

↳ to make them all positive, i.e. so deviations ABOVE and BELOW the expected value don't cancel each other out.

why not $|x - E(x)|$? There are other transformations to make the distances positive, eg absolute value. Why square them? Nothing fundamentally special about squaring them, it just ends up being mathematically convenient in some of the downstream uses of variance (whereas absolute value doesn't behave as nicely).

In addition to variance, we also like to talk about standard deviation, which is the sqrt of variance

$$\sigma_x = \text{standard deviation}(X) = \sqrt{\text{var}(X)}$$

It's the same information as the variance, but has the same units as the RV itself.

for variance, because of the squared differences $(x-E(X))^2$, variance is in the RV's units, squared. So for the distrib. of # kittens in a litter, the variance describes the spread in units of kittens^2, while std dev is in units of kittens, which may be more useful to think about.

For the 2 coin flips example, Y = # tails, $E(Y) = 1$,

$$\text{var}(Y) = (0-1)^2 P(Y=0) + (1-1)^2 P(Y=1) + (2-1)^2 P(Y=2)$$
$$= 1 \cdot 1/4 + 0 + 1 \cdot 1/4 = 1/2$$

– joint distributions –

Sometimes we're interested in studying more than 1 RV at the same time.

Suppose your cat is going to have a kitten, and you're interested in what color it will be (X) and whether it's a boy or girl (Y).

new kitten
X = fur color
Y = sex

we could consider those two variables separately and see:

y	M	F
p(y)	0.5	0.5

x	orange	black	multi
p(x)	5/8	1/8	2/8

← (Imagine we know mom & dad's fur colors, and genetics dictates these probabilities)

but each possible PAIR of X and Y has its own probability. For instance, we could have:

	M	F
orange	3/8	2/8
black	1/8	0
multi	0	2/8

(note: the individual distributions of X and Y do not give you enough info to fill out this table (unless they're independent - more on that later). I'm giving you this joint distribution)

These are all the possible PAIRS of outcomes. Their probabilities have to add up to 1.

The set of all possible PAIRS of outcomes & their probabilities is the joint distribution of X and Y

A related concept is "conditional probability"

What is the probability of the kitten being orange, given that it's a boy?

Look at the "Male" column of the joint dist. table.

↳ Within the set of outcomes w/ a male kitten, the probability of orange is 3 times the prob. of black, and those are the only possible outcomes.

$$P(X = \text{orange} \mid Y = M) = \frac{P(\text{orange} \;\&\; Y = M)}{P(Y = M)} = \frac{3/8}{3/8 + 1/8} = 3/4$$

$P(Y = M)$ ↳ equals the sum of the probabilities in the $Y = M$ table

we calculate the conditional probability by dividing the probability of an orange male kitten by the total probability of a male kitten.

↳ conceptually: we restricted ourselves to the set of outcomes where the kitten is male, then rescaled those probabilities for each color so that they add up to 1. (Given that the kitten is male, it has to be SOME color.)

Notice that we said $P(\text{orange}) = 5/8$ at the beginning. If we add up all the probabilities of outcomes with an orange cat in the joint dist. table (ie the "orange" row), it equals $5/8$.

So the probability of the kitten being orange is different depending on sex

$$P(\text{orange}) \neq P(\text{orange} \mid \text{Male})$$

If the distribution of one RV is different depending on the value of the other RV, we call them "dependent"

dependent : $P(X|Y) \neq P(X)$

the alternative is for them to be INdependent

X, Y independent : $P(X|Y) = P(X)$

In our example, independence would mean that the probability of orange fur would be the same for male and female kittens.

For independent RVs, it's also true that the joint probability of X and Y equals the product of each individual probability

X, Y independent : $P(X=x, Y=y) = P(X=x)\,P(Y=y)$

For the kittens example,

$P(\text{orange}, M) = 3/8$

$P(\text{orange}) = 5/8$

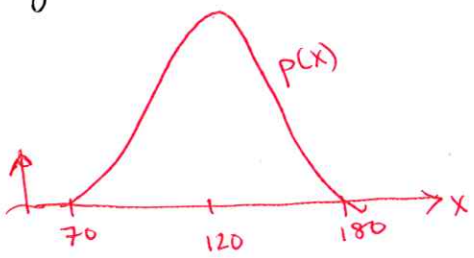
$P(M) = 1/2$

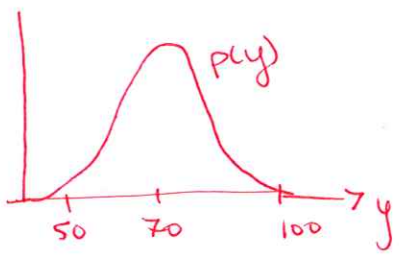
so $\underbrace{P(\text{orange}, M)}_{3/8} \neq \underbrace{P(\text{orange})}_{5/8} * \underbrace{P(M)}_{1/2}$

the joint probability does NOT factor, & and X, Y are NOT independent in this example (i.e. are dependent).

That example used categorical variables. The same idea applies for numerical random variables that are discrete or continuous.

eg: Suppose X = blood pressure (BP) and Y = resting heart rate (RHR)





If X and Y are independent, someone with a low RHR and someone with a high RHR are equally likely to have elevated blood pressure.

If X and Y are dependent, we aren't claiming that one causes the other, just that there is an association. (dependence ≠ causality)

To describe the association, we can consider its direction and its strength.

associations can be positive or negative in direction -

* "positive" would mean that people with high BP tend to have high RHR
* "negative" would mean that people with high BP tend to have LOW RHR

To quantify the strength of the association, we use covariance and correlation.

The covariance is defined as

$$\mathbb{E}\left[(X - \mathbb{E}(X)) * (Y - \mathbb{E}(Y)) \right]$$

which equals

$$\sum_{\text{all } (x,y) \text{ pairs}} (x - \mathbb{E}(X)) * (y - \mathbb{E}(Y)) \, \mathbb{P}(X = x, Y = y)$$

$\mathbb{P}(X = x, Y = y)$: joint probability

Why does this measure what we want?

If X and Y are positively associated, X and Y tend to be large at the same time.

X, Y both large $\mapsto$ $x - \mathbb{E}(X) > 0$ & $y - \mathbb{E}(Y) > 0$

so $(x - \mathbb{E}(X)) * (y - \mathbb{E}(Y)) > 0$

X, Y positive association also means that they tend to be SMALL at the same time

x, y both small $\longrightarrow$ $x - \mathbb{E}(X) < 0$ & $y - \mathbb{E}(Y) < 0$

so $(x - \mathbb{E}(X)) * (y - \mathbb{E}(Y)) > 0$

For a very strong positive association, pairs of values (x, y) that are *both* large or *both* small are more likely, so the joint probability $P(X = x, Y = y)$ is large for those pairs.

↳ Overall, the covariance is a sum of mostly big positive numbers.

For a weak association, sometimes X and Y are simultaneously large, but sometimes only one of them is, making $(x - \mathbb{E}(X)) * (y - \mathbb{E}(Y)) < 0$. In this case, the covariance becomes a sum of some positive and some negative terms that mostly cancel each other out, resulting in covariance (X,Y) being close to 0.

If X and Y are independent, then the covariance is EXACTLY zero.

X, Y independent $\Rightarrow$ covariance(X, Y) = 0

Correlation is a related quantity, that just rescales the covariance to be in the range [-1, 1]

$$\text{correlation}(X, Y) = \frac{\text{covariance}(X,Y)}{\sigma_x \sigma_y} = \rho_{xy}$$

σ_x: standard deviation of X

σ_y: standard deviation of Y

Sometimes we use the letter "ρ" to refer to correlation.

Correlation has important consequences for how sums of random variables behave.

Suppose we want to add 2 random variables together. What is the expected value of the sum?

$\mathbb{E}(X+Y) = \mathbb{E}(X) + \mathbb{E}(Y)$ regardless of dependence/independence

Regardless of the correlation between X and Y, the expected value of their sum equals the sum of their individual expected values.

Imagine X is a student's score on the math section of the SAT exam, and Y is their score on the reading section.

$\mathbb{E}(X) = 500$, $\mathbb{E}(Y) = 500$ $\Rightarrow$ $\mathbb{E}(X+Y) = 500 + 500 = 1000$

The expected value of the combined score is the sum of the expected score on each section.

This is not always true for variances!

If a student's scores on the 2 sections are independent, the variances add.

$\text{var}(X) = 100^2$, $\text{var}(Y) = 100^2$ } if X, Y indep, then $\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$

$= 100^2 + 100^2$

But if the section scores are correlated, we CAN'T just add them. If they're correlated, the variance of their sum ALSO depends on the correlation ρ_{xy}.

$$\text{var}(X+Y) = \text{var}(X) + \text{var}(Y) + 2\rho_{xy}\sigma_x\sigma_y$$

(ρ_{xy}: correlation; σ_x: standard deviation of X; σ_y: standard deviation of Y)

If scores on the 2 sections are positively correlated, total scores are more spread out.

↳ someone above average on math is also likely to be above average on reading, pushing their total score further away from 1000 ($1000 = \mathbb{E}(X+Y)$)

If scores on the 2 sections are negatively correlated, total scores are less spread out

↳ someone above avg on math is likely to be below average on reading, cancelling out & pushing their total score back toward 1000